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MAT251_TW | Chapter 7: Sequences and Series

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Question **1**

Complete

Not graded

For which formula does $\{r_j\} = \{2, 4, 8, 16, 32, \dots\}$?

Select one:

- a. $r_j = 2j$
- b. $r_j = 2^{j+1}$
- c. $r_j = 2 + j$
- d. $r_j = 2^j$

Plugging $j = 1, 2, 3, 4,$ and 5 into the formula $r_j = 2^j$, results in the correct elements:

$r_1 = 2^1 = 2,$

$r_2 = 2^2 = 4,$

$r_3 = 2^3 = 8,$

$r_4 = 2^4 = 16,$

$r_5 = 2^5 = 32.$

The correct answer is: $r_j = 2^j$

Question **2**

Complete

Not graded

Evaluate $\lim_{n\to\infty} \frac{sin(n^2+1)\frac{\pi}{2}}{n}$.

Select one:

- a. 0
- b. It does not exist.
- c. $\frac{1}{2}$
- d. 1

Because $-1 \le \sin(n^2+1)\frac{\pi}{2} \le 1$,

$$\lim_{n\to\infty} \frac{\sin(n^2+1)\frac{\pi}{2}}{n} = 0.$$

The correct answer is: 0

Question **3**

Complete

Not graded

Which of the following is the limit of $\left\{ \frac{\ln n}{n} \right\}$?

Select one:

- a. ∞
- b. 1
- c. 0
- d. −∞

If you try to plug in $n = \infty$, we get ∞/∞.
This is an indeterminate form, so you
can use L'Hopital's rule.

$$\lim_{n\to\infty} \frac{\ln n}{n} = \lim_{n\to\infty} \frac{1/n}{1} = \lim_{n\to\infty} \frac{1}{n} = 0$$

This demonstrates that n grows faster
than $\ln n$.

The correct answer is: 0

Question **4**

Complete

Not graded

Evaluate the following as true or false.

The sequence { x_n }, where $x_n = cosn\pi$, converges.

Select one:

- True
- False

$\cos 0 = \cos 2\pi = \cos 4\pi = \cos 6\pi = \ldots = 1,$
so whenever n is even, $x_n = 1$.
$\cos \pi = \cos 3\pi = \cos 5\pi = \cos 7\pi = \ldots = -1,$
so whenever n is odd, $x_n = -1$.
In other words, the sequence
alternates between 1 and -1, but it
never approaches a single, unique
value. Therefore, the sequence
diverges.

The correct answer is 'False'.

Question **5**

Complete

Not graded

For which of the following formulas does $\{d_n\} = \{2, 6, 12, 20, 30, \ldots\}$?

Select one:

- a. $d_n = 4n - 2$
- b. $d_n = 2n$
- c. $d_n = n^2 + 3n + 2$
- d. $d_n = n^2 + n$

$1^2 + 1 = 2 = d_1$

$2^2 + 2 = 6 = d_2$

$3^2 + 3 = 12 = d_3$

$4^2 + 4 = 20 = d_4$

$5^2 + 5 = 30 = d_5$

The correct answer is: $d_n = n^2 + n$

Question **6**

Complete

Not graded

Which of the following is the limit of $\left\{ ln\left(\frac{n+1}{n} \right) \right\}$?

Select one:

- a. 0
- b. 1
- c. $ln(2)$
- d. ∞

Rewrite the fraction inside the logarithm.

$$\frac{n+1}{n} = 1 + \frac{1}{n}$$

Take the limit of this fraction as n approaches infinity.

$$\lim_{n \to \infty} \left(1 + \frac{1}{n} \right) = 1.$$

Because ln is a continuous function,

$$\lim_{n \to \infty} \ln \frac{n+1}{n} = \ln \lim_{n \to \infty} \frac{n+1}{n} = \ln 1 = 0.$$

The correct answer is: 0

Question **7**

Complete

Not graded

Which of the following is the 20th term of the sequence { a_n }, where $a_n = n + 2$?

Select one:

- a. 20
- b. 22
- c. 40
- d. 18

The 20th term of the sequence $a_n = n + 2$ is $a_{20} = 20 + 2 = 22$.

The correct answer is: 22

Question **8**

Complete

Not graded

What is the 20th term of the sequence $\{q_j\}$, where $q_j = (-1)^j$?

Select one:

- a. −1
- b. 0
- c. 1

$(-1)^{20} = 1$. Notice that all the terms with an odd index are −1 and all the terms with an even index are 1.

The correct answer is: 1

Question **9**

Complete

Not graded

Which of the following is the limit of $\left\{ n^2 e^{-n} \right\}$?

Select one:

- a. e
- b. 1
- c. 0
- d. ∞

Because $e^{-n} = \frac{1}{e^n}$, rewrite

$n^2 e^{-n} = \frac{n^2}{e^n}$. If you try to plug in

$n = \infty$, you get the indeterminate form

$\frac{\infty}{\infty}$. Because you have an indeterminate

form, we can use L'Hopital's rule.

$$\lim_{n\to\infty} \frac{n^2}{e^n} = \lim_{n\to\infty} \frac{2n}{e^n} = \lim_{n\to\infty} \frac{2}{e^n} = 0,$$

because $e^n \to \infty$ as $n \to \infty$.

The correct answer is: 0

Question **10**

Complete

Not graded

Evaluate $\lim_{n\to\infty} \frac{n^2+3n}{2n^2-n}$.

Select one:

- a. $\frac{1}{3}$
- b. 1
- c. $\frac{1}{2}$
- d. $\frac{3}{2}$

$$\lim_{n\to\infty} \frac{n^2+3n}{2n^2-n} = \lim_{n\to\infty} \frac{1+\frac{3}{n}}{2-\frac{1}{n}}$$

$$= \frac{1}{2}$$

The correct answer is: $\frac{1}{2}$

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